Monday

1. HW 10.2 Due tonight
2. Test 3 Next Monday
   A. Study Guide already on CourseDen
3. Rose Curves
   A. number of petals
   B. converting to parametric
   C. graphing
4. Start 10.4 Polar Area

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3. $x = 7t \quad y = t^9$ tangent at $t=-1$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{9t^8}{7}\Bigg|_{t=-1} = \frac{9}{7}$$

point when $t=-1$:

$$x = 7(-1) = -7 \qquad y - y_1 = m(x - x_1)$$

$$y = (-1)^9 = -1 \qquad y + 1 = \frac{9}{7}(x+7)$$

Length of one arch of $x = 3(t - \sin t)$

$y = 3(1 - \cos t)$

$$\int_0^{2\pi} dS \qquad dS = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

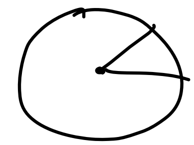
$$= \sqrt{(3(1-\cos t))^2 + (3(\sin t))^2}\, dt$$

$$= \sqrt{9(1 - 2\cos t + \cos^2 t) + 9\sin^2 t}\, dt$$

$$= 3\sqrt{1-2\cos t + \underbrace{\cos^2 t + \sin^2 t}_{1}}\,dt = 3\sqrt{2-2\cos t}\,dt$$

$$\text{Length} = 3\int_0^{2\pi}\sqrt{2-2\cos t}\,dt = 24$$

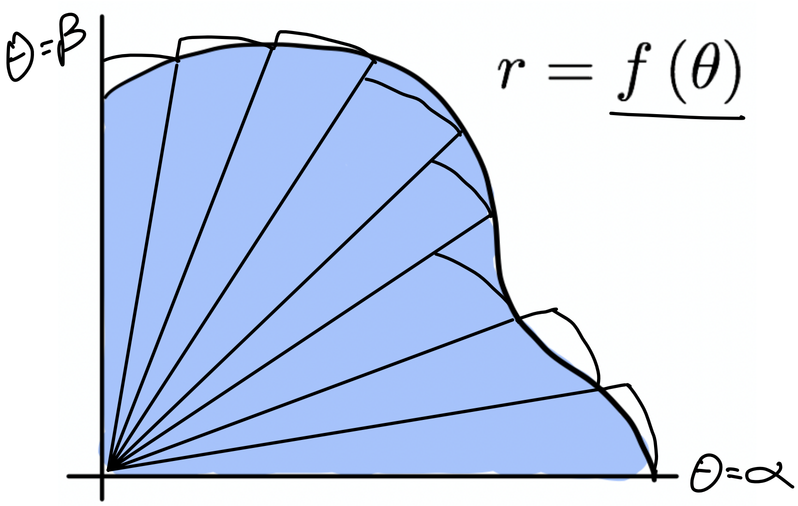
10.4 Areas in Polar Coordinates



Recall:

- The area of a Circle with radius r is $A = \pi r^2$.
- The area of a sector of a circle with central angle θ (in radians) is $A_{\text{sector}} = \frac{1}{2}r^2\theta$

We use the sector area formula to find the area of an irregular curve:

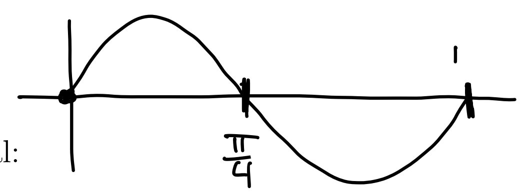


Area inside $r = f(\theta)$:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2 d\theta$$

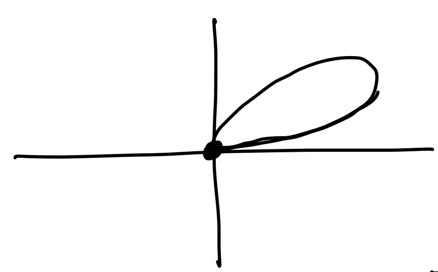
Rose Curves: A function of the form $r = \sin(a\theta)$ or $r = \cos(a\theta)$ is called a **rose curve**. If a is even, the curve has $2a$ petals; if a is odd, it has a petals.

0.0.1 Example:



The rose curve $r = \sin 4\theta$ has 8 petals. Find the area of one petal:

$$A = \frac{1}{2}\int_0^{\pi/4} (\sin 4\theta)^2 d\theta = \frac{1}{2}\int_0^{\pi/4} \sin^2 4\theta \, d\theta$$



$$= \frac{1}{2}\int_0^{\pi/4} \frac{1}{2}\left[1 - \cos 8\theta\right] d\theta$$

$\sin^2 x = \frac{1}{2}\left[1 - \cos 2x\right]$

$$= \frac{1}{4}\left[\theta - \frac{1}{8}\sin 8\theta\right]_0^{\pi/4} = \frac{1}{4}\left[\frac{\pi}{4} - \frac{1}{8}\underbrace{\sin 2\pi}_{0}\right] - \frac{1}{4}\left[0 - \frac{1}{8}\underset{0}{\sin 0}\right]$$

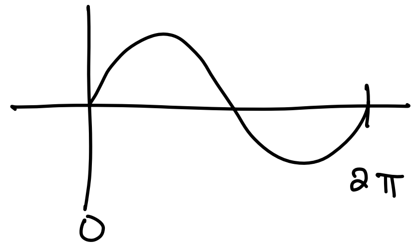
$$= \boxed{\frac{\pi}{16}}$$

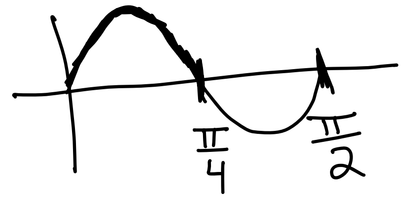
$\sin \underbrace{4\theta}$

$0 \le 4\theta \le 2\pi$

$0 \le \theta \le \frac{2\pi}{4}$

$0 \le \theta \le \frac{\pi}{2}$



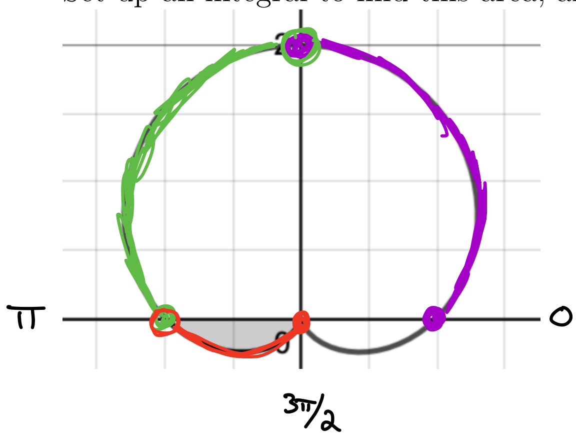


$y = \sin 4\theta$

0.0.2 Example:

$\frac{\pi}{2}$

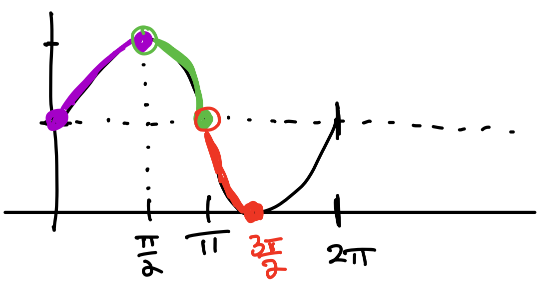
Set up an integral to find this area, and use a calculator to evaluate it: $r = 1 + \sin\theta$



$$A = \frac{1}{2}\int_{\pi}^{3\pi/2} (1 + \sin\theta)^2 \, d\theta$$

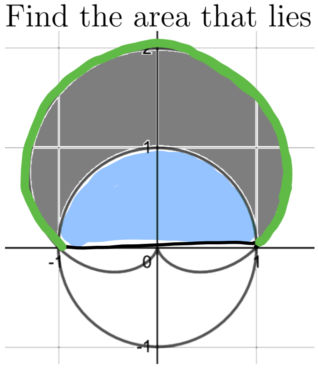
$$= .178097\ldots.$$

$y = 1 + \sin x$



0.0.3 Example:

Find the area that lies outside of $r = 1$ and inside of $1 + \sin\theta$.

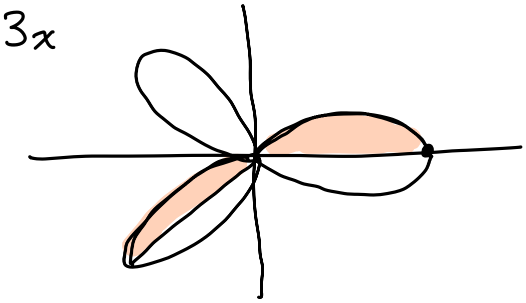


Area = Area of cardiod from 0 to π − Area of semicircle,

$$\frac{1}{2}\int_0^{\pi} (1+\sin\theta)^2 \, d\theta - \frac{\pi(1)^2}{2}$$

$$= 4.356 - \frac{\pi}{2} = \boxed{2.785}$$

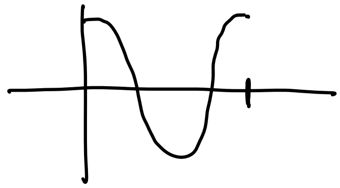
Extra Example: $r = 4\cos 3x$



Find area of one petal.

$$\frac{1}{2}\int_0^{\pi/3} (4\cos 3\theta)^2 \, d\theta$$

$$\frac{1}{2}\cdot 16 \int_0^{\pi/3} \cos^2 3\theta \, d\theta$$



$y = 4\cos 3x$

$0 \le 3x \le 2\pi$

$0 \le x \le \frac{2\pi}{3}$

$$8\int_0^{\pi/3} \frac{1}{2}\left[1 + \cos 6\theta\right] d\theta$$

$$= 4\left[\theta + \frac{1}{6}\sin 6\theta\right]_0^{\pi/3} = 4\left[\frac{\pi}{3} + \frac{1}{6}\sin 2\pi\right] = \boxed{\frac{4\pi}{3}}$$